1. Data Analysis Python Project - Blinkit Analysis

1.0.1 Import Liabraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

1.0.2 Import Raw Data

```
df = pd.read_csv('blinkit_data.csv')
```

1.0.3 Sample Data

```
df.head(10)
```

Columns 1–8 of 13

	Item Fat Content	Item Identifier	Item Type	Outlet Establishment Year	Outlet Identifier	Outlet Location Type	Outlet Size
0	Regular	FDX32	Fruits and Vegetables	2012	OUT049	Tier 1	Medium
1	Low Fat	NCB42	Health and Hygiene	2022	OUT018	Tier 3	Medium
2	Regular	FDR28	Frozen Foods	2010	OUT046	Tier 1	Small
3	Regular	FDL50	Canned	2000	OUT013	Tier 3	High
4	Low Fat	DRI25	Soft Drinks	2015	OUT045	Tier 2	Small
5	low fat	FDS52	Frozen Foods	2020	OUT017	Tier 2	Small
6	Low Fat	NCU05	Health and Hygiene	2011	OUT010	Tier 3	Small
7	Low Fat	NCD30	Household	2015	OUT045	Tier 2	Small
8	Low Fat	FDW20	Fruits and Vegetables	2000	OUT013	Tier 3	High
9	Low Fat	FDX25	Canned	1998	OUT027	Tier 3	Medium

Columns 9–13 of 13 · first column repeated

	Outlet Type	Item Visibility	Item Weight	Sales	Rating
0	Supermarket Type1	0.100014	15.10	145.4786	5.0
1	Supermarket Type2	0.008596	11.80	115.3492	5.0
2	Supermarket Type1	0.025896	13.85	165.0210	5.0
3	Supermarket Type1	0.042278	12.15	126.5046	5.0
4	Supermarket Type1	0.033970	19.60	55.1614	5.0
5	Supermarket Type1	0.005505	8.89	102.4016	5.0
6	Grocery Store	0.098312	11.80	81.4618	5.0
7	Supermarket Type1	0.026904	19.70	96.0726	5.0

	Outlet Type	Item Visibility	Item Weight	Sales	Rating
8	Supermarket Type1	0.024129	20.75	124.1730	5.0
9	Supermarket Type3	0.101562	NaN	181.9292	5.0

```
df.tail()
```

Columns 1–8 of 13

	Item Fat Content	Item Identifier	Item Type	Outlet Establishment Year	Outlet Identifier	Outlet Location Type	Outlet Size
8518	low fat	NCT53	Health and Hygiene	1998	OUT027	Tier 3	Medium
8519	low fat	FDN09	Snack Foods	1998	OUT027	Tier 3	Medium
8520	low fat	DRE13	Soft Drinks	1998	OUT027	Tier 3	Medium
8521	reg	FDT50	Dairy	1998	OUT027	Tier 3	Medium
8522	reg	FDM58	Snack Foods	1998	OUT027	Tier 3	Medium

Columns 9–13 of 13 · first column repeated

	Outlet Type	Item Visibility	Item Weight	Sales	Rating
8518	Supermarket Type3	0.000000	NaN	164.5526	4.0
8519	Supermarket Type3	0.034706	NaN	241.6828	4.0
8520	Supermarket Type3	0.027571	NaN	86.6198	4.0
8521	Supermarket Type3	0.107715	NaN	97.8752	4.0
8522	Supermarket Type3	0.000000	NaN	112.2544	4.0

1.0.4 Size of Data

```
print("Size of Data:", df.shape)
```

```
Size of Data: (8523, 12)
```

1.0.5 Field Info

```
df.columns
```

```
Index(['Item Fat Content', 'Item Identifier', 'Item Type',
       'Outlet Establishment Year', 'Outlet Identifier',
       'Outlet Location Type', 'Outlet Size', 'Outlet Type', 'Item Visibility',
       'Item Weight', 'Sales', 'Rating'],
      dtype='str')
```

1.0.6 Data Types

```
df.dtypes
```

```
Item Fat Content                str
Item Identifier                 str
Item Type                       str
Outlet Establishment Year     int64
Outlet Identifier               str
Outlet Location Type            str
Outlet Size                     str
Outlet Type                     str
Item Visibility             float64
Item Weight                 float64
Sales                       float64
Rating                      float64
dtype: object
```

1.0.7 Data Cleaning

```
print(df['Item Fat Content'].unique())
```

```
<ArrowStringArray>
['Regular', 'Low Fat', 'low fat', 'LF', 'reg']
Length: 5, dtype: str
```

```
df['Item Fat Content'] = df['Item Fat Content'].replace({'LF' : 'Low Fat', 'low fat'
: 'Low Fat', 'reg' : 'Regular' })
```

```
print(df['Item Fat Content'].unique())
```

```
<ArrowStringArray>
['Regular', 'Low Fat']
Length: 2, dtype: str
```

1.0.8 Business Requirements

KPI'S Requirements

Total Sales

```
# Total Sales:
total_sales = df['Sales'].sum()

# Avg Sales:
avg_sales = df['Sales'].mean()

# No. of Items Sold:
no_of_items_sold = df['Sales'].count()

#Avg Ratings:
avg_ratings = df['Rating'].mean()

print(f"Total Sales:${total_sales :,.0f}")
print(f"Avgerage Sales:${avg_sales :,.0f}")
print(f"No. of Items Sold:{no_of_items_sold :,.0f}")
print(f"Avg Ratings:{avg_ratings :,.0f}")
```

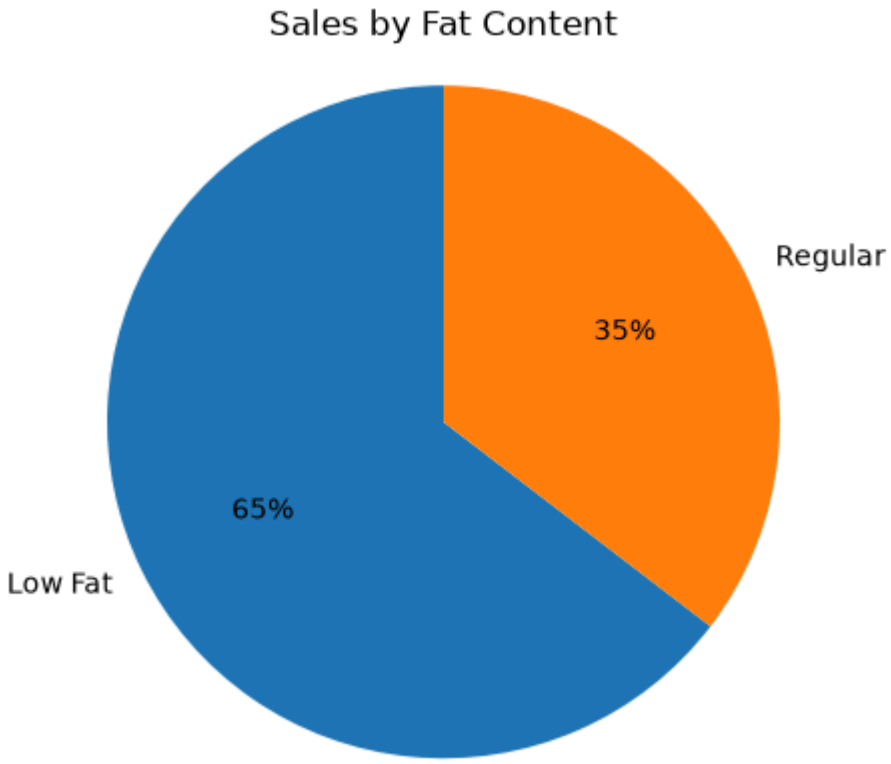
```
Total Sales:$1,201,681
Avgerage Sales:$141
No. of Items Sold:8,523
Avg Ratings:4
```

1.1 Charts Requirements

1.1.9 Total Sales by Fat Content

```
sales_by_fat_content = df.groupby('Item Fat Content')['Sales'].sum()

plt.pie(sales_by_fat_content, labels = sales_by_fat_content.index, autopct =
'%.0f%%', startangle = 90)
plt.title("Sales by Fat Content")
plt.axis('equal')
plt.show()
```



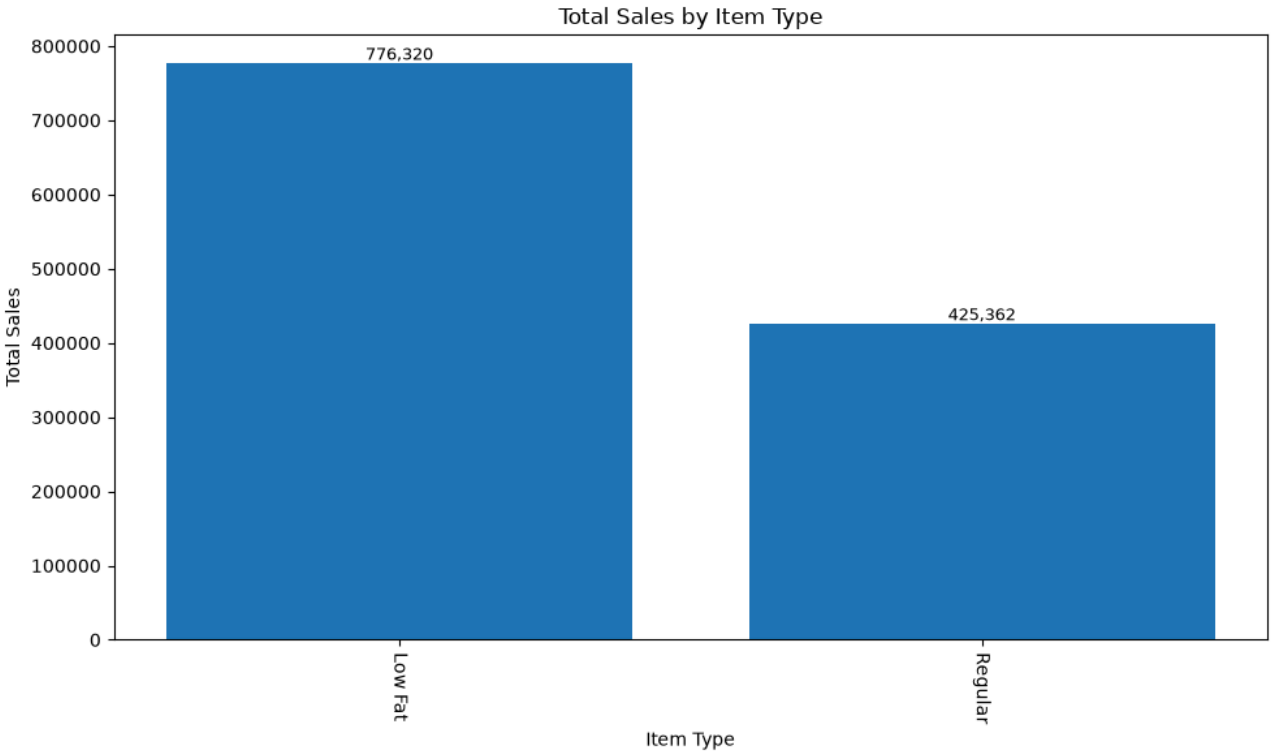
1.1.10 Total Sales by Item Type

```
sales_by_type = df.groupby('Item Fat Content')['Sales'].sum().sort_values(ascending =
False)

plt.figure(figsize = (10,6))
bars = plt.bar(sales_by_type.index, sales_by_type.values)
plt.xticks(rotation = -90)
plt.xlabel('Item Type')
plt.ylabel('Total Sales')
plt.title('Total Sales by Item Type')

for bar in bars:
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height(),
             f'{bar.get_height():,.0f}', ha = 'center', va = 'bottom', fontsize = 9)

plt.tight_layout()
plt.show()
```

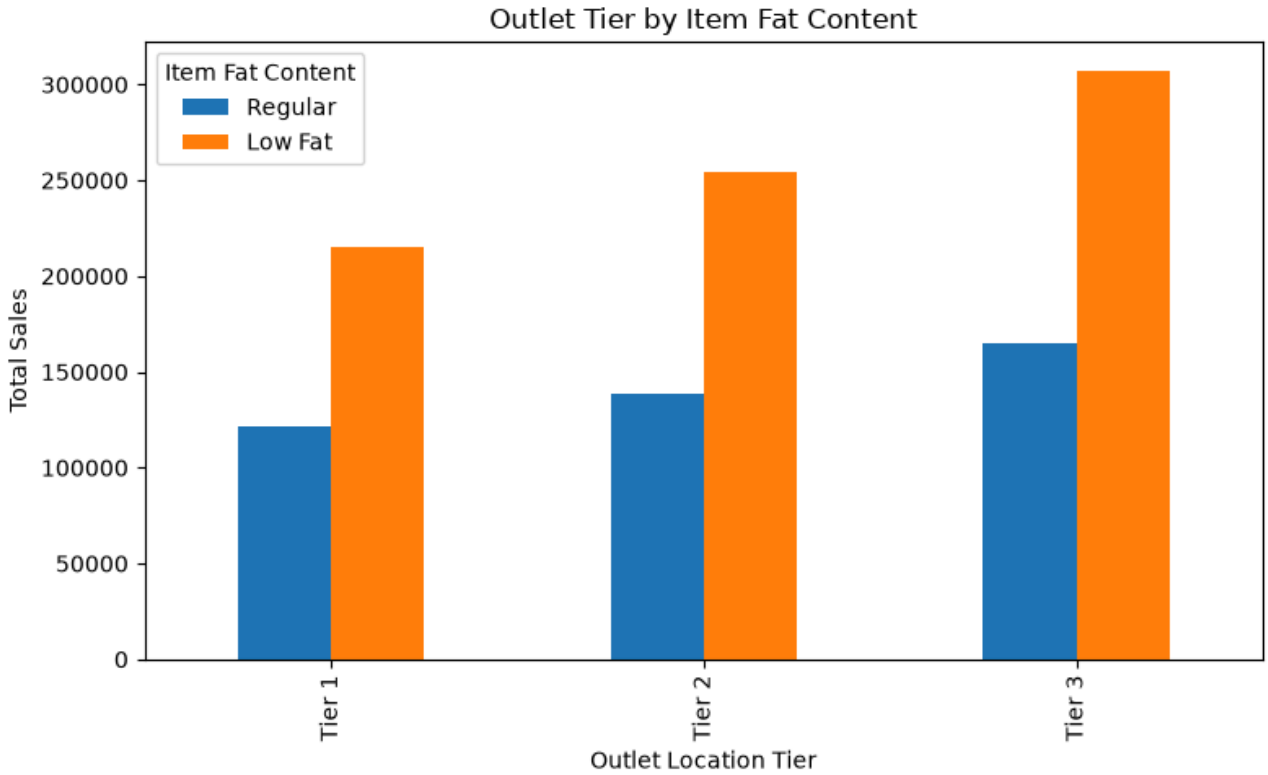


1.1.11 Fat Content by Outlet for Total Sales

```
grouped = df.groupby(['Outlet Location Type', 'Item Fat Content'])
['Sales'].sum().unstack()
grouped = grouped[['Regular', 'Low Fat']]

ax = grouped.plot(kind = 'bar', figsize = (8,5), title = 'Outlet Tier by Item Fat
Content')

plt.xlabel('Outlet Location Tier')
plt.ylabel('Total Sales')
plt.legend(title = 'Item Fat Content')
plt.tight_layout()
plt.show()
```



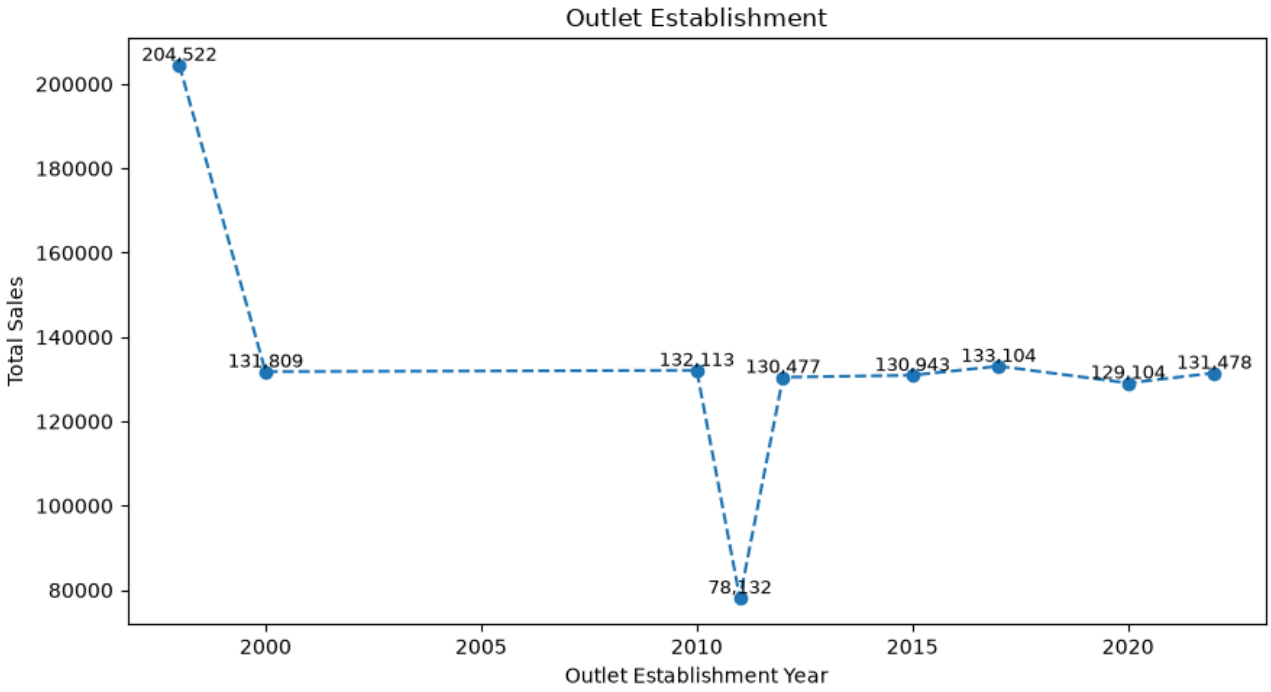
1.1.12 Total Sales by Outlet Establishment

```
sales_by_year = df.groupby('Outlet Establishment Year')['Sales'].sum().sort_index()

plt.figure(figsize = (9,5))
plt.plot(sales_by_year.index, sales_by_year.values, marker = 'o', linestyle = '--')
plt.xlabel('Outlet Establishment Year')
plt.ylabel('Total Sales')
plt.title('Outlet Establishment')

for x,y in zip(sales_by_year.index, sales_by_year.values):
    plt.text(x,y,f'{y:,.0f}', ha = 'center', va = 'bottom', fontsize = 9)

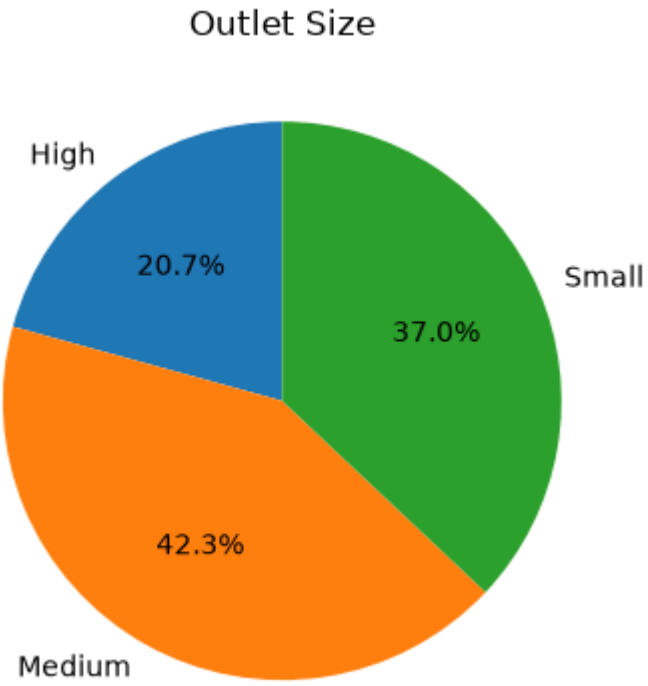
plt.tight_layout()
plt.show()
```



1.1.13 Sales by Outlet Size

```
sales_by_size = df.groupby('Outlet Size')['Sales'].sum()

plt.figure(figsize = (4,4))
plt.pie(sales_by_size, labels = sales_by_size.index, autopct = '%1.1f%%', startangle
= 90)
plt.title('Outlet Size')
plt.tight_layout()
plt.show()
```



1.1.14 Sales by Outlet Location

```
sales_by_location = df.groupby('Outlet Location Type')['Sales'].sum().reset_index()
sales_by_location = sales_by_location.sort_values('Sales', ascending = False)

plt.figure(figsize = (8,3)) # Smaller height enough width
ax = sns.barplot(x = 'Sales', y = 'Outlet Location Type', data = sales_by_location)

plt.title('Total Sales by Outlet Location Type')
plt.xlabel('Total Sales')
plt.ylabel('Outlet Location Type')

plt.tight_layout() # Ensures layout fits without scroll
plt.show()
```

